



CBU5000 V210 Radar Driver (2.1)

RADAR TEST APP USER MANUAL

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1 Introduction

This document gives details of the current radar driver release and the accompanying test app. The test app demonstrates two modes of operation – continuous-sensing mode, and frame-based mode. The primary purpose of the app is to demonstrate the use of the radar driver for different frame format, retrieve the CIR data and send over the UART.

2 Test App Parameters

The list of radar parameters are given below. In the command-line version of the test app, twelve parameters must be provided. Table 1 shows the table of the parameters, with the details given in the subsequent subsections.

Table 1. Parameters for the test app

idx	Radar Config Parameter, P[idx]	Description
0	mode_PRF	0 – BPRF 1 – HPRF, 2 – LG4A
1	power_code	Power code 0-60
2	scale_bit	Receiver scale bit. Recommended values: 4 – LG4A 6 – BPRF and HPRF
3	gain_idx	0-7 Receiver gain index
4	CIR_tap_start	Starting index. Minimum value is 0. CIR_Tap_start = 30 is the assumed R = 0 m.
5	CIR_tap_end	Ending index. Total CIR taps returned is (CIR_Tap_end – CIR_Tap_start + 1) CIR_tap_end maximum value is 235 corresponding to the maximum range R = 30.8 m. Each tap is approximately 15.024 cm.
6	sensing_mode	0 – if continuous mode 1 – if frame-based mode
7	sensing_burst_num	1-256 but the actual value to be calculated at the back-end
8	sensing_interval_ms	Interval of the CIR in slow-time. Minimum value is 1. Maximum value is 10000 (10 seconds).
9	frame_interval_ms	Group CIRS into one frame. One frame has sensing and idle parts. <i>Used if sensing_mode is 1.</i>
10	num_cir_per_frame	Number of CIRS in a single frame. Maximum is 32. <i>Used if sensing_mode is 1.</i>
11	deep_sleep_en	1 or 0 . Puts the SoC in deep-sleep during the idle part. <i>Used if sensing_mode is 1.</i>
12	num_rx_mode	1 - to return Rx2 CIR only, 2 - to return both Rx1 and Rx2
13	preamble_index	Preamble code selection 1-8 LG4A 9-24 BPRF 25-32 HPRF

2.1 Mode_PRF

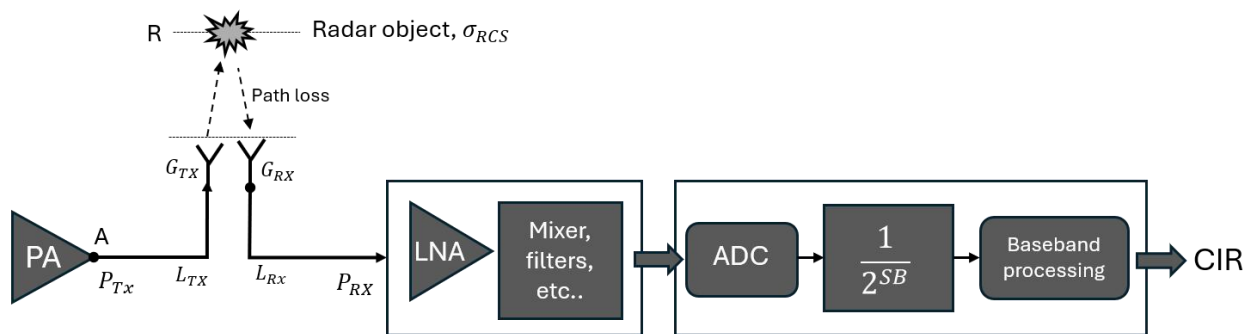
The radar driver supports 3 modes with different pulse repetition frequencies – BPRF, HPRF, and legacy LG4A.

Table 2. Different PRF modes used in radar sensing

mode_PRF	Mode	Packet Frame Length	Transmit Power Range	Mean PRF
0	BPRF	73.2us (127x4x2ns)x72	-55 to -25 dBm/MHz	62.89 MHz
1	HPRF	54.5us (91x4x2ns)x72	-52.5 to -22.5 dBm/MHz	111.09 MHz
2	LG4A	71.4us (31x16x2ns)x72	-61 to -31 dBm/MHz	16.1 MHz

2.2 Power Code, Gain Index and Scale Bit

The power code refers to the transmitter power amplifier gain that controls the output power. Every step in the power code corresponds to around 0.5 dB. The gain index controls the receiver gain with each increment equal to approximately -6 dB. The scale bit is a pre-scaler in the baseband processing.



Power code controls the gain of the power amplifier. Range of values: 0-62.

Gain index controls the gain of the receiver chain. It also affects the effective noise figure. Range of values: 0-7, 3 bits

Scale bit adds post-scaling 2^{SB} after ADC sampling (pre-scaling before baseband processing) Range of values for SB : 0-7, 3bits

Figure 2-1. Illustration for the components being controlled by the power code, gain index and scale bit.

Table 3. Power code and corresponding power density

Power code	Power @ Chip Port BPRF (dBm/MHz)	Power @ Chip Port HPRF (dBm/MHz)	Power @ Chip Port LG4A (dBm/MHz)
0	-55	-52.5	-61
1	-54.5	-52	-60.5
2	-54	-51.5	-60
3	-53.5	-51	-59.5
8-12	Unused		
58	-26	-23.5	-20
59	-25.5	-23	-19.5
60	-25	-22.5	-19

Table 4. Gain index and the corresponding gain, noise figure and input power saturation point

Gain Index	LNA Gain (dB)	Noise Figure (dB)	Input Saturation Point (dBm, BPRF)	Input Saturation Point (dBm, HPRF)	Input Saturation Point (dBm, LG4A)
0	60	4.0	-70	-67.5	-76
1	55	4.1	-65	-62.5	-71
2	50	4.4	-60	-57.5	-66
3	44	5.8	-54	-51.5	-60
4	37	8.6	-48	-45.5	-54
5	31	12.1	-41	-38.5	-47
6	23	20.0	-33	-30.5	-39
7	12	28.9	-22	-19.5	-28

2.3 CIR_tap_start and CIR_tap_end

The **CIR_tap_start** and **CIR_tap_end** refers to the first and last index of the requested CIR IQ data. **Tap index = 30** is the approximate location of R=0 m, estimated from the SMA port of the evaluation kit. Tap index = 0 corresponds to some *negative* distance.

2.4 Continuous sensing (sensing_mode = 0)

When **sensing_mode** is set to 0, the radar is continuously transmitting and acquiring CIR data. Each CIR will be available at regular intervals equal to the **sensing_interval_ms** provided. This mode is mainly used for simultaneous sensing and processing.

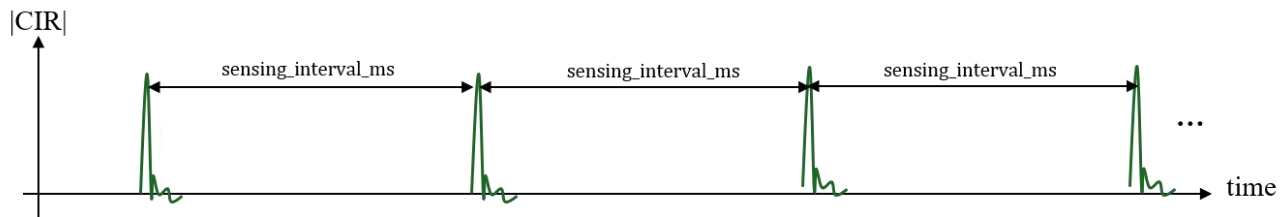


Figure 2-2. Continuous sensing mode

2.5 Burst mode sensing_burst_num

The test app also allows for burst mode wherein several CIRs can be combined by averaging. This is enabled when **sensing_burst_num** is set to a value greater than 1. This effectively maintains the signal magnitude and reduces the noise floor (increase in signal-to-noise-ratio).

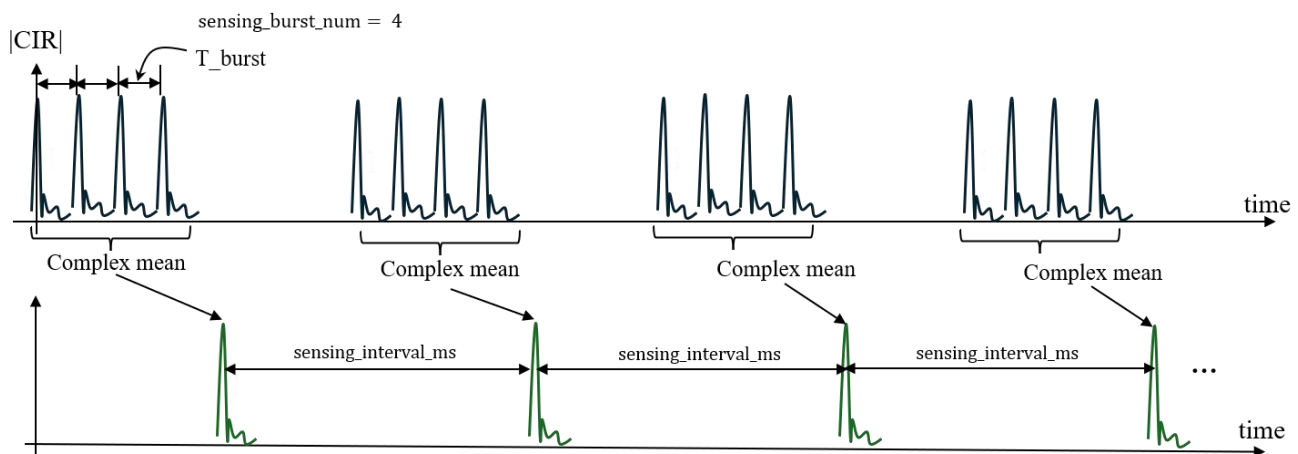


Figure 2-3. Continuous sensing mode with burst.

In the burst mode, the inter-burst time T_{burst} changes with the number of receivers and length of taps need to be returned. Approximately,

$$T_{burst_us} \approx 104 + (28 + 0.57 \cdot N) \cdot num_rx_mode$$

where N is the number of CIR taps ($CIR_{tap_end} - CIR_{tap_start} + 1$) and T_{burst} is in μs .

Note that the total burst time must not exceed the desired sensing interval:

$$sensing_interval_ms > sensing_burst_num \cdot \left(\frac{T_{burst_us}}{1000} \right).$$

Table 5. Maximum number of sensing_burst_num for num_rx_mode=1

Number of Taps	sensing_burst_num (max)
$N \leq 17$	7 * sensing_interval_ms
$18 \leq N \leq 59$	6 * sensing_interval_ms
$60 \leq N \leq 119$	5 * sensing_interval_ms
$120 \leq N \leq 207$	4 * sensing_interval_ms
$208 \leq N \leq 241$	3 * sensing_interval_ms

Table 6. Maximum number of sensing_burst_num for num_rx_mode=2

Number of Taps	sensing_burst_num (max)
$0 < N \leq 5$	6 * sensing_interval_ms
$6 \leq N \leq 35$	5 * sensing_interval_ms
$36 \leq N \leq 78$	4 * sensing_interval_ms
$79 \leq N \leq 151$	3 * sensing_interval_ms
$152 \leq N \leq 241$	2 * sensing_interval_ms

2.6 Frame-based sensing (sensing_mode = 1)

In frame-based sensing, several CIRs are collected to form a single radar frame. This is enabled by setting the **sensing_mode = 1**. If enabled, frame-based parameters **frame_interval_ms**, **num_cir_per_frame** are used. The **sensing_interval_ms** and **sensing_burst_num** are also applicable in this mode. If needed, the driver is also capable of deep sleep by enabling **deep_sleep_en** and is only used in this frame-based mode.

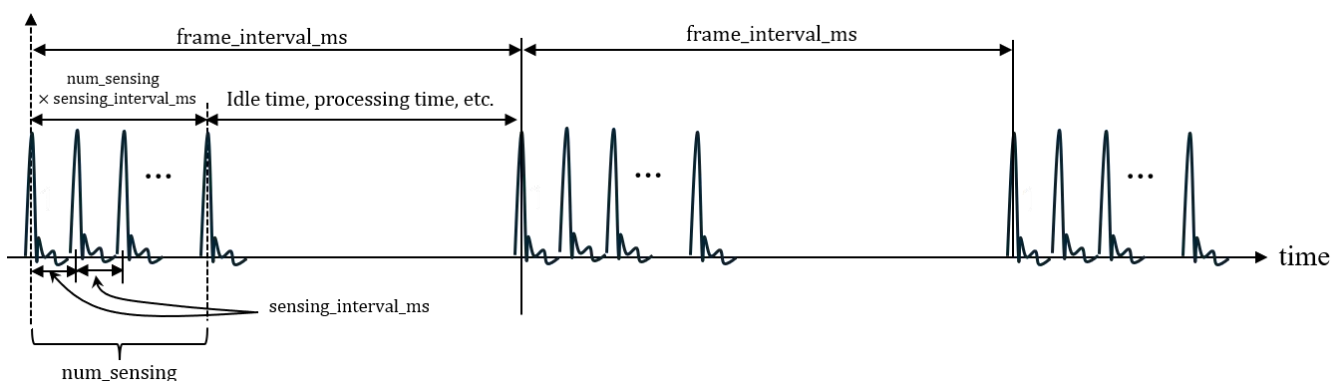


Figure 2-4. Frame-based mode. Burst mode can be enabled but not shown in this figure.

3 Test APP Data Format

One purpose of the test app is to return the CIR data via UART for algorithm development. This section gives details on the timing diagram describing the sequence from data acquisition to CIR data printing. Information about the binary format is also provided to facilitate parsing.

3.1 Sensing mode = 0 (continuous sensing)

In sensing mode = 0, the CIR data is sent after one CIR data is available. An example of parameters settings is shown in Table 7. The corresponding timing diagram is shown in Figure 3- 1.

Table 7. An example of parameter settings for continuous mode

idx	Radar Config Parameter	Value
0	mode_PRF	0 (BPRF)
1	power_code	48
2	scale_bit	6
3	gain_idx	2
4	CIR_tap_start	30 (R=0 m)
5	CIR_tap_end	70 (R=6m)
6	sensing_mode	0 (continuous)
7	sensing_burst_num	4
8	sensing_interval_ms	5
9	frame_interval_ms	100 (don't care since sensing_mode = 0)
10	num_cir_per_frame	32 (don't care since sensing_mode = 0)
11	deep_sleep_en	0 (don't care since sensing_mode = 0)
12	num_rx_mode	2 (Rx1 and Rx2)
13	preamble_index	9
UART Command	f,0,48,6,2,30,70,0,4,5,100,32,0,2,9	

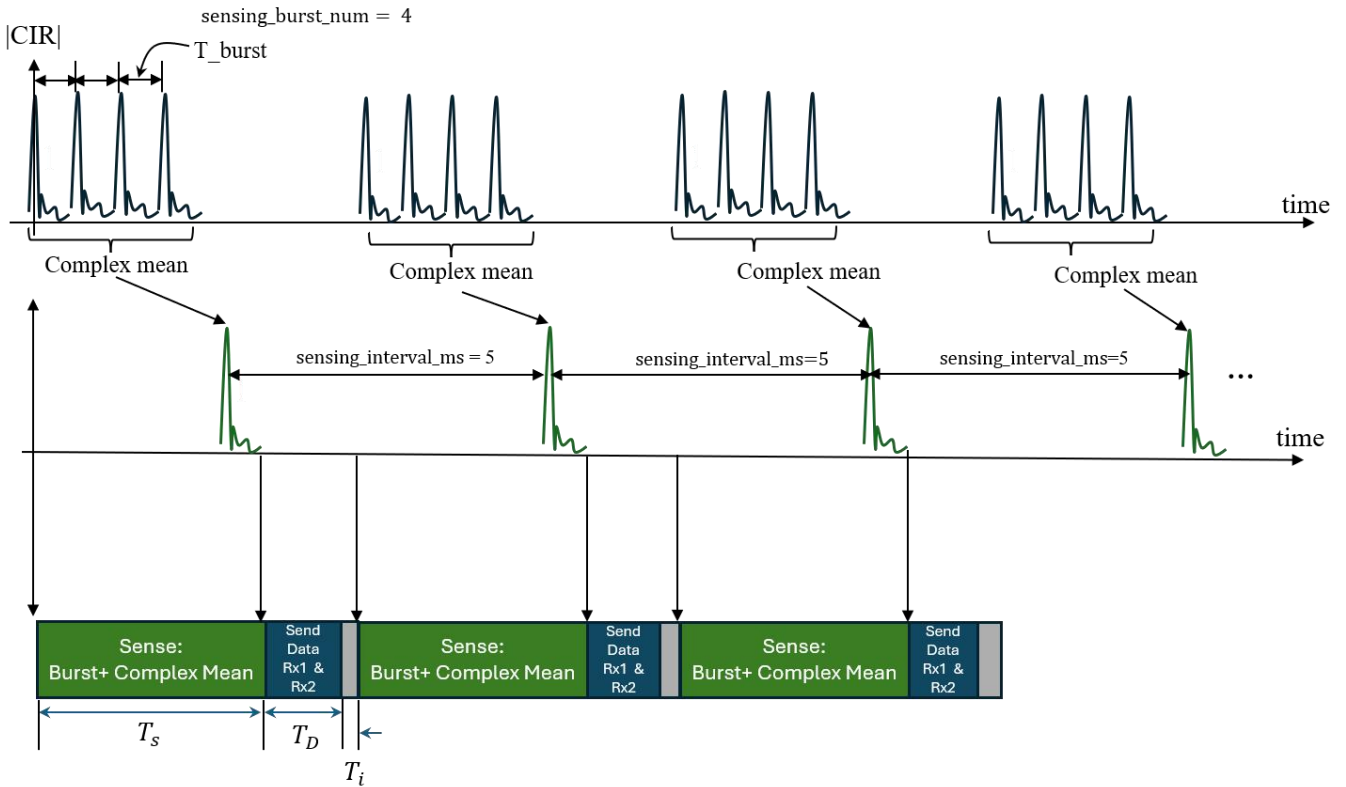


Figure 3-1. Test app timing diagram for continuous mode. T_s , T_D and T_i are the sensing, data transfer and idle time for each CIR cycle, respectively.

It requires T_s to complete all the bursts and the subsequent averaging.

$$T_s(\text{ms}) = \text{sensing_burst_num} \cdot \left(\frac{T_{\text{burst_us}}}{1000} \right).$$

The UART printout requires $33 \mu\text{s}$ for one tap (32-bit complex) per receiver at 921600 baud rate.

$$T_D(\text{ms}) \cong \frac{0.033 \text{ ms}}{\text{tap} \cdot \text{receiver}} \cdot (\text{CIR}_{\text{tap_end}} - \text{CIR}_{\text{tap_start}} + 1) \cdot \text{num_rx_mode}$$

For long CIR sequence, the sensing interval must be sufficiently long to allow the data to be sent via UART.

$$\text{sensing_interval_ms} > T_s + T_D$$

3.2 Sensing mode = 1 (frame-based sensing)

For frame-based sensing, the data is sent after every frame. An example of parameter settings is shown in Table 8. The corresponding timing diagram are shown in Figure 3-2.

Table 8. An example of parameter settings for frame-based mode

idx	Radar Config Parameter, P[idx]	Value
0	mode_PRF	0 (BPRF)
1	power_code	48
2	scale_bit	6
3	gain_idx	2
4	CIR_tap_start	30 (R=0 m)
5	CIR_tap_end	70 (R=6m)

6	sensing_mode	1 (frame-based)
7	sensing_burst_num	1
8	sensing_interval_ms	1
9	frame_interval_ms	100
10	num_cir_per_frame	32
11	deep_sleep_en	0
12	num_rx_mode	2
13	preamble_index	9
UART Command	f,0,48,6,2,30,70,1,1,1,100,32,0,2,9	

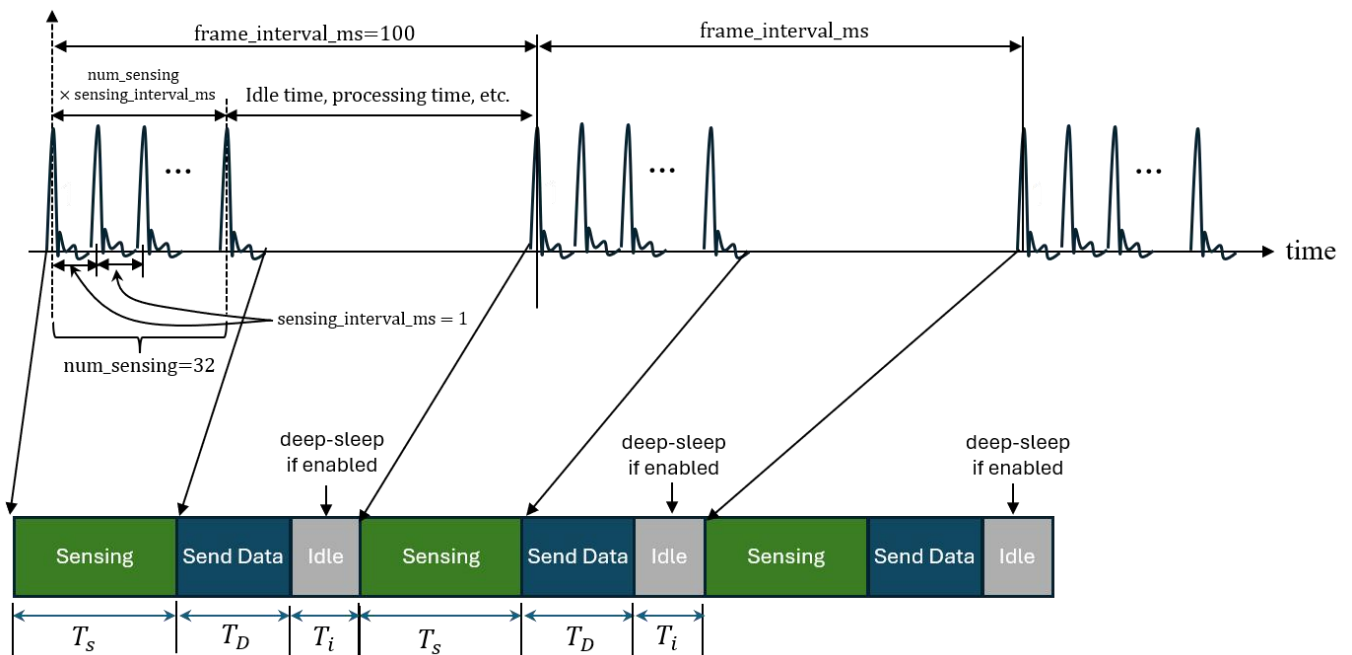


Figure 3-2. Test app timing diagram for frame-based mode. T_s , T_D and T_i are the sensing, data transfer and idle time for one complete frame, respectively.

For long CIR sequence, the frame interval must be sufficiently long to allow the data to be sent via UART at 921600 baud rate. Note that the data is sent only after one frame is complete. The sensing and data transfer time are calculated as follows:

$$T_s (\text{ms}) \cong \text{num_sensing} \times \text{sensing_interval_ms}$$

$$T_D (\text{ms}) \cong \frac{0.033 \text{ ms}}{\text{tap} \cdot \text{receiver}} \cdot (\text{CIR}_{\text{tap_end}} - \text{CIR}_{\text{tap_start}} + 1) \cdot \text{num_sensing} \cdot \text{num_rx_mode}$$

$$\text{frame_interval_ms} > T_s + T_D$$

Note that burst-mode is still applicable in this sensing mode, hence the sensing interval should still satisfy:

$$\text{sensing_interval_ms} > \text{sensing_burst_num} \cdot \left(\frac{T_{\text{burst_us}}}{1000} \right).$$

3.3 Logging in binary format

The test app prints the headers in ASCII followed by a binary stream of the IQ data. An example shown in Figure 3-1 is the CIR data returned by the test app via UART.

New File 1	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	
00000	0A	6D	6F	64	65	5F	50	52	46	20	76	61	6C	75	65	3A	mode_PRF value:
00010	20	32	0A	70	6F	77	65	72	5F	63	6F	64	65	20	76	61	2 power_code va
00020	6C	75	65	3A	20	31	38	0A	73	63	61	6C	65	5F	62	69	lue: 18 scale_bi
00030	74	20	76	61	6C	75	65	3A	20	36	0A	67	61	69	6E	5F	t value: 6 gain_
00040	69	64	78	20	76	61	6C	75	65	3A	20	34	0A	43	49	52	idx value: 4 CIR
00050	5F	74	61	70	5F	73	74	61	72	74	20	76	61	6C	75	65	_tap_start value
00060	3A	20	32	38	0A	43	49	52	5F	74	61	70	5F	65	6E	64	: 28 CIR_tap_end
00070	20	76	61	6C	75	65	3A	20	31	30	38	0A	73	65	6E	73	value: 108 sens
00080	69	6E	67	5F	6D	6F	64	65	20	76	61	6C	75	65	3A	20	ing_mode value:
00090	31	0A	73	65	6E	73	69	6E	67	5F	62	75	72	73	74	5F	l sensing_burst_
000A0	6E	75	6D	20	76	61	6C	75	65	3A	20	31	0A	73	65	6E	num value: 1 sen
000B0	73	69	6E	67	5F	69	6E	74	65	72	76	61	6C	5F	6D	73	sing_interval_ms
000C0	20	76	61	6C	75	65	3A	20	32	0A	66	72	61	6D	65	5F	value: 2 frame_
000D0	69	6E	74	65	72	76	61	6C	5F	6D	73	20	76	61	6C	75	interval_ms valu
000E0	65	3A	20	32	35	30	0A	6E	75	6D	5F	63	69	72	5F	70	e: 250 num_cir_p
000F0	65	72	5F	66	72	61	6D	65	20	76	61	6C	75	65	3A	20	er_frame value:
00100	33	32	0A	64	65	65	70	5F	73	6C	65	65	70	5F	65	6E	32 deep_sleep_en
00110	20	76	61	6C	75	65	3A	20	30	0A	6E	75	6D	5F	72	78	value: 0 num_rx
00120	5F	6D	6F	64	65	20	76	61	6C	75	65	3A	20	32	0A	70	_mode value: 2 p
00130	72	65	61	6D	62	6C	65	5F	69	6E	64	65	78	20	76	61	reamble_index va
00140	6C	75	65	3A	20	31	0A	49	6E	64	65	78	52	65	66	20	lue: 1 IndexRef
00150	76	61	6C	75	65	3A	20	33	30	0A	0A	3E	41	50	50	5F	value: 30 >APP_
00160	52	61	64	61	72	53	74	61	72	74	0A	FF	FF	00	FF	FF	RadarStart 00 00
00170	FF	00	00	00	01	32	13	4C	07	38	FA	EC	FB	00	FF	00	0 00000000 0
00180	FF	BC	FF	44	00	00	00	48	FF	FC	FF	B8	FF	FC	00	40	0000 H000000 0
00190	00	48	00	3C	00	3C	00	3C	00	00	00	00	00	00	FF	C0	H < < < 00
001A0	FE	FC	FE	BC	FF	9C	00	7C	00	68	00	20	00	00	00	00	000000 h
001B0	00	00	FF	FC	FF	FC	00	00	00	00	FF	FC	FF	F8	FF	FC	0000 000000
001C0	00	00	00	00	00	00	00	00	FF	FC	00	00	00	00	00	00	00
001D0	00	00	00	00	00	00	FF	FC	00	00	00	00	00	00	00	00	00
001E0	FF	FC	FF	FC	00	00	00	FF	FC	00	00	00	00	00	FF	FC	0000 00 00
001F0	FF	FC	00	00	00	00	FF	F4	00	00	FF	FC	FF	FC	FF	FC	00 00 000000
00200	FF	F4	FF	FC	FF	FC	FF	FC	00	00	00	00	00	00	00	00	00000000
00210	00	00	FF	FC	FF	FC	00	00	F1	A0	F9	7C	02	98	03	88	0000 000 0000
00220	00	80	FF	7C	FF	FC	00	40	00	40	00	3C	00	3C	00	3C	00 00 0 0 < < <
00230	00	00	00	00	00	00	00	10	00	1C	00	00	00	00	00	20	0 0
00240	FF	FC	FF	74	FF	7C	00	00	00	40	00	10	00	20	00	08	00000 0 0 0
00250	00	08	00	00	00	14	00	40	00	40	00	3C	00	00	FF	FC	0 0 0 0 < 00
00260	00	00	00	00	00	00	00	00	00	20	00	00	00	00	00	00	00
00270	00	00	00	00	00	00	00	00	00	04	00	00	00	00	00	00	0
00280	00	00	00	20	00	0C	00	00	00	00	00	00	00	00	00	00	↑
00290	00	00	00	00	00	08	00	24	00	24	00	2C	FF	FC	00	00	0 \$ \$,00
002A0	00	10	00	38	00	00	00	00	00	00	00	00	00	00	00	00	0 8
002B0	00	00	00	00	00	00	00	00	00	FF	3C	FF	BC	00	3C		0<00 <
002C0	FF	BC	FE	44	FD	84	FE	C0	00	BC	00	BC	00	58	00	3C	00000000 0 0 X <
002D0	00	40	00	00	FF	B8	FF	C0	00	00	FF	F8	FF	C8	FF	F4	0 0000 000000
002E0	FF	DC	FF	98	FF	40	FF	40	FF	D4	00	40	00	40	00	80	0000000000 0 0 0
002F0	00	80	00	08	FF	C4	FF	C4	FF	C8	FF	F4	FF	FC	00	00	0 0000000000

Figure 3-3. CIR data returned by the test App via UART.

The header is immediately followed by the CIR data. The CIR data is composed of the following:

- num_rx_mode = 1: Identifier, Sequence and Frame ID, Rx2 Data

FF FF 00FF FFFF00	XX XX XX	YY	I2	Q2 ...
Identifier 7 bytes	Sequence ID 3 bytes	Frame ID 1 byte	Rx2 I and Q data I: N * 2bytes Q: N * 2bytes N is number of taps	

- num rx mode = 2: Identifier, Sequence and Frame ID, Rx1 Data, Rx2 Data

FF FF 00FF FFFF00	XX XX XX	YY	I1	Q1 ...	I2	Q2 ...
Identifier 7 bytes	Sequence ID 3 bytes	Frame ID 1 byte	Rx1 I and Q data I: N * 2bytes Q: N * 2bytes N is number of taps	Rx2 I and Q data I: N * 2bytes Q: N * 2bytes N is number of taps		

For continuous mode (sensing_mode=0), only the sequence ID is incremented for every CIR printed. For frame-based mode (sensing_mode=1), the frame ID is incremented whenever a new frame is printed. The sequence ID is incremented for each CIR and CIRs within the same frame have the same frame ID. The sequence ID is not being reset for each new frame.

3.4 Test App Command

Primarily, the command-line interface version of the test app accepts two commands:

- Command to start the radar logging requires 14 parameters in the sequence as indicated in Table 1. For example:

“f.0.48.6.2.38.58.1.1.1.1000.4.0.2.9”

The screenshot shows a terminal window titled "COM6 - Tera Term VT". The menu bar includes File, Edit, Setup, Control, Window, and Help. The main area displays the following commands and their values:

```
>f,0,48,6,2,30,50,1,1,1,1000,4,0,2,9
mode_PRF value: 0
power_code value: 48
scale_bit value: 6
gain_idx value: 2
CIR_tap_start value: 30
CIR_tap_end value: 50
sensing_mode value: 1
sensing_burst_num value: 1
sensing_interval_ms value: 1
frame_interval_ms value: 1000
num_cir_per_frame value: 4
deep_sleep_en value: 0
num_rx_mode value: 2
preamble_index value: 9
IndexRef value: 30
```

Below this, the command ">APP_RadarStart" is entered, followed by a series of garbled characters and symbols, likely representing a corrupted or incomplete response from the device.

- Command to stop the acquisition

“f.o”

- **IMPORTANT** For deep sleep mode, “f,0” has to be input repeatedly and continuously as the command has to be sent while the SoC is ‘awake’. Alternatively, a GPIO (GPIO 7 in the release) can be set to low to prevent the hardware from going into deep-sleep – then followed by the stop “f,0” command.

4 Revision History

Version No	Description	Date
0.1	Document Creation	14 Aug 2026
0.2	Change title, update formula	19 Aug 2026
0.3	Add preamble code selection for BPRF and HPRF	20 Aug 2026
0.4	- Add preamble selection for LG4A - Update burst time based on 64 MHz clock - Shift the reference to index =30 (R=0)	25 Aug 2026
0.5	Fix formatting issues	31 Aug 2026

5 Disclaimers

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